

Laboratório 3

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```
library(readr)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
library(leaflet)
library(ggplot2)
library(maps)
```

1

Airports

```
airports <- read_csv(
  "airports.csv",
  col_types = cols_only(
    IATA_CODE = col_character(),
    CITY      = col_character(),
```

```

    STATE = col_character(),
    LATITUDE = col_double(),
    LONGITUDE = col_double()
  )
)
airports

```

```

# A tibble: 322 x 5
  IATA_CODE CITY      STATE LATITUDE LONGITUDE
  <chr>      <chr>      <chr>   <dbl>   <dbl>
1 ABE      Allentown    PA      40.7    -75.4
2 ABI      Abilene        TX      32.4    -99.7
3 ABQ      Albuquerque    NM      35.0   -107.
4 ABR      Aberdeen       SD      45.4    -98.4
5 ABY      Albany         GA      31.5    -84.2
6 ACK      Nantucket       MA      41.3    -70.1
7 ACT      Waco           TX      31.6    -97.2
8 ACV      Arcata/Eureka   CA      41.0   -124.
9 ACY      Atlantic City   NJ      39.5    -74.6
10 ADK      Adak            AK      51.9   -177.
# i 312 more rows

```

Airlines

```
airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

-- Column specification -----

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airlines
```

```

# A tibble: 14 x 2
  IATA_CODE AIRLINE
  <chr>      <chr>

```

1	UA	United Air Lines Inc.
2	AA	American Airlines Inc.
3	US	US Airways Inc.
4	F9	Frontier Airlines Inc.
5	B6	JetBlue Airways
6	OO	Skywest Airlines Inc.
7	AS	Alaska Airlines Inc.
8	NK	Spirit Air Lines
9	WN	Southwest Airlines Co.
10	DL	Delta Air Lines Inc.
11	EV	Atlantic Southeast Airlines
12	HA	Hawaiian Airlines Inc.
13	MQ	American Eagle Airlines Inc.
14	VX	Virgin America

flights

```
# lista que vai acumular as estatísticas suficientes de cada bloco
lista_stats <- list()
processa_bloco <- function(bloco, pos) {
  stats_bloco <- bloco |>
    filter(substr(DESTINATION_AIRPORT, 1, 1) != "1") |>
    drop_na() |>
    group_by(DESTINATION_AIRPORT) |>
    summarise(
      soma_atraso = sum(ARRIVAL_DELAY),
      n_voos      = n(),
      .groups = "drop"
    )
  # acumula na lista
  lista_stats[[length(lista_stats) + 1]] <- stats_bloco
}
read_csv_chunked(
  "flights.csv",
  callback = SideEffectChunkCallback$new(processa_bloco),
  chunk_size = 1e6,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY       = col_double()
  )
)
```

NULL

```
# combina as estatísticas suficientes de todos os blocos
flights <- bind_rows(lista_stats) |>
  group_by(DESTINATION_AIRPORT) |>
  summarise(
    soma_atraso = sum(soma_atraso),
    n_voos      = sum(n_voos),
    .groups = "drop"
  ) |>
  mutate(atraso_medio = soma_atraso / n_voos)
flights
```

```
# A tibble: 322 x 4
  DESTINATION_AIRPORT soma_atraso n_voos atraso_medio
  <chr>                <dbl>   <int>         <dbl>
1 ABE                  12874    2220          5.80
2 ABI                   9077    2224          4.08
3 ABQ                106415   18953          5.61
4 ABR                 -2227     657         -3.39
5 ABY                   7501     864          8.68
6 ACK                   2393     487          4.91
7 ACT                 16392    1541         10.6
8 ACV                 12259    1267          9.68
9 ACY                 41744    3530         11.8
10 ADK                 -284      89         -3.19
# i 312 more rows
```

2. Join

Aeroportos em flights mas ausentes em airports

```
aeroportos_ausentes <- flights |>
  anti_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
aeroportos_ausentes
```

```
# A tibble: 0 x 4
# i 4 variables: DESTINATION_AIRPORT <chr>, soma_atraso <dbl>, n_voos <int>,
#   atraso_medio <dbl>
```

```
flights <- flights |>
  left_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
flights
```

A tibble: 322 x 8

	DESTINATION_AIRPORT	soma_atraso	n_voos	atraso_medio	CITY	STATE	LATITUDE
	<chr>	<dbl>	<int>	<dbl>	<chr>	<chr>	<dbl>
1	ABE	12874	2220	5.80	Allentown	PA	40.7
2	ABI	9077	2224	4.08	Abilene	TX	32.4
3	ABQ	106415	18953	5.61	Albuquerque~	NM	35.0
4	ABR	-2227	657	-3.39	Aberdeen	SD	45.4
5	ABY	7501	864	8.68	Albany	GA	31.5
6	ACK	2393	487	4.91	Nantucket	MA	41.3
7	ACT	16392	1541	10.6	Waco	TX	31.6
8	ACV	12259	1267	9.68	Arcata/Eu~	CA	41.0
9	ACY	41744	3530	11.8	Atlantic ~	NJ	39.5
10	ADK	-284	89	-3.19	Adak	AK	51.9

i 312 more rows

i 1 more variable: LONGITUDE <dbl>

3.aeropertos por estado

```
aeropertos_por_estado <- airports |>
  count(STATE, name = "n_aeropertos") |>
  arrange(desc(n_aeropertos))
aeropertos_por_estado
```

A tibble: 54 x 2

	STATE	n_aeropertos
	<chr>	<int>
1	TX	24
2	CA	22
3	AK	19
4	FL	17
5	MI	15
6	NY	14
7	CO	10
8	MN	8
9	MT	8

```
10 NC 8
# i 44 more rows
```

4. Mapa dos atrasos por aeroporto

```
this <- flights |>
  filter(!is.na(LATITUDE), !is.na(LONGITUDE)) |>
  leaflet() |>
  addTiles() |>
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      DESTINATION_AIRPORT, " - ", CITY, "/", STATE,
      "<br>Atraso médio: ", round(atraso_medio, 1), " min"
    )
  )
```

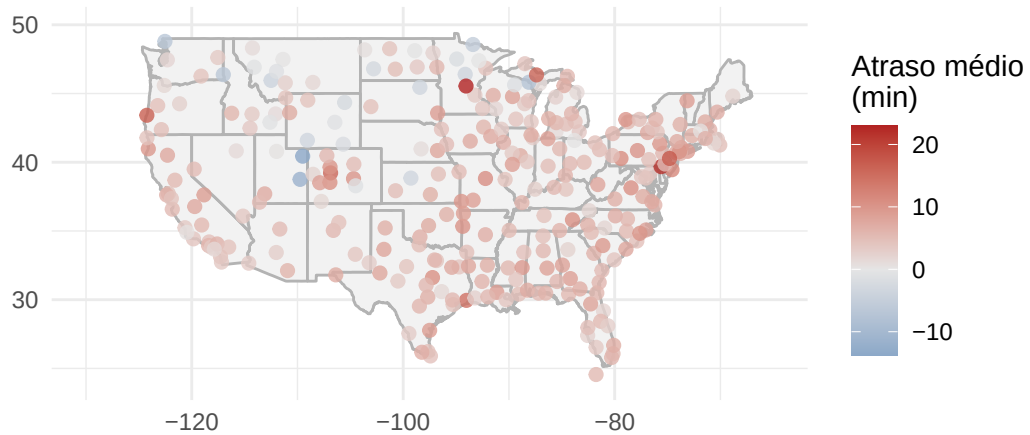
```
mapa_dados <- flights |>
  filter(!is.na(LATITUDE), !is.na(LONGITUDE))

estados <- map_data("state")

ggplot() +
  geom_polygon(
    data = estados,
    aes(x = long, y = lat, group = group),
    fill = "grey95", color = "grey70"
  ) +
  geom_point(
    data = mapa_dados,
    aes(x = LONGITUDE, y = LATITUDE, color = atraso_medio),
    size = 2, alpha = 0.8
  ) +
  scale_color_gradient2(
    low = "steelblue", mid = "grey90", high = "firebrick",
    midpoint = 0, name = "Atraso médio\n(min)"
  ) +
  coord_fixed(1.3, xlim = c(-130, -65), ylim = c(24, 50)) +
  labs(
    title = "Atraso médio de chegada por aeroporto de destino",
```

```
x = NULL, y = NULL
) +
theme_minimal()
```

Atraso médio de chegada por aeroporto de destino



```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-04 21:21:00 -03"
```